

Reviewer Pack — Coxeter-Diagram Invariant Bounds on Communication Complexity

Entry 0a741fca781a · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Coxeter-Diagram Invariant Bounds on Communication Complexity

Entry ID: 0a741fca781a **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-31 03:57:59 UTC

1. Conjecture

Field A (mathematical branch): Combinatorial Group Theory (specifically, Coxeter diagrams) **Field B** (complexity object): Communication Complexity

Statement:

For any given boolean function f with n inputs and communication complexity $C(f)$, the number of non-identity relations in the corresponding Coxeter diagram of f is bounded by a function of $C(f)$ and $\log(n)$, i.e., $|R(f)| \leq \alpha(C(f), \log(n))$, where $|R(f)|$ is the number of relations in the Coxeter diagram and α is an absolute constant.

Rationale (proposer's reasoning):

Coxeter diagrams encode the symmetry properties of combinatorial objects, which are known to relate to communication complexity. A deep connection between Coxeter diagrams and communication complexity could provide new insights into the nature of computational tasks requiring communication.

Taxonomy category: coxeter-diagram-communication-complexity (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 6594f1e5eae34cb4

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all generated boolean functions with communication complexity $C(f)$, the ratio $|R(f)|/\log(n)$ does not exceed an absolute constant $\alpha(C(f), \log(n))$ across all seeds. The criterion is falsified if any seed produces a ratio exceeding $\alpha(C(f), \log(n))$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	1.00	SAFE	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	1.00	SAFE	SAFE
KARP_LIPTON	SAFE	1.00	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Coxeter-diagram invariant" AND "communication complexity" - "Coxeter group theory" AND "bounds on communication complexity" - "boolean function" AND "Coxeter diagram relations"

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
from fractions import Fraction
import math
import sys

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def communication_complexity(f):
        n = len(f)
        if n == 1:
            return 0
        for i in range(1, n):
            if all(f[j] != f[j + i] for j in range(n - i)):
                return i
        return n

    def coxeter_diagram_relations(f):
        n = len(f)
        relations = set()
        for i in range(n):
            for j in range(i + 1, n):
                if f[i] != f[j]:
                    relations.add((i, j))
        return relations

    n_values = [5, 10, 15, 20, 30, 40]
    total_ratio = 0
    instances_tested = 0
    n_max = 0
```

```

for n in n_values:
    f = generate_boolean_function(n)
    C_f = communication_complexity(f)
    R_f = coxeter_diagram_relations(f)

    if len(R_f) == 0 or C_f == 0:
        continue

    log_n = math.log2(n)
    ratio = Fraction(len(R_f), log_n) if log_n != 0 else float('inf')
    total_ratio += ratio
    instances_tested += 1
    n_max = max(n_max, n)

if instances_tested == 0:
    return {
        "metric_name": "ratio",
        "metric_value": float('nan'),
        "instances_tested": 0,
        "n_max": n_max,
        "conjecture_holds": False,
        "counterexample": "No valid instances found"
    }

mean_ratio = total_ratio / instances_tested
return {
    "metric_name": "ratio",
    "metric_value": float(mean_ratio),
    "instances_tested": instances_tested,
    "n_max": n_max,
    "conjecture_holds": True,
    "counterexample": ""
}

if __name__ == "__main__":
    if len(sys.argv) > 1:
        seeds = [int(s) for s in sys.argv[1:]]
    else:
        primes = [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97]
        seeds = primes[:30]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_ratio = sum(r["metric_value"] for r in results if not math.isnan(r["metric_value"])) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(math.isnan(r["metric_value"]) for r in results):
        print("RESULT: INCONCLUSIVE no_valid_instances")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_ratio} std=nan support_fraction={support_fraction}")
    else:

```

```

first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
print(f"RESULT: FALSIFIED counterexample=\"ratio_exceeds_bound\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_58c1655c.py", line 90, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_58c1655c.py", line 56, in run_trial
    ratio = Fraction(len(R_f), log_n) if log_n != 0 else float('inf')
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/fractions.py", line 277, in __new__
    raise TypeError("both arguments should be "
TypeError: both arguments should be Rational instances

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which prevents a definitive evaluation of the conjecture. | next: Review and debug the test code to ensure it can complete without crashing and produce results for further analysis.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	15056
2	preregistration	ollama_remote	glm4:latest	0	0	21028
3	novelty	ollama_remote	glm4:latest	0	0	12490
4	novelty	ollama_remote	glm4:latest	0	0	25708
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	19449
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	18840
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15568
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15472
9	judge	ollama_remote	glm4:latest	0	0	14186

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 157796 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/0a741fca781a.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/0a741fca781a.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/0a741fca781a.tar.gz` (if generated)